UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

a) How classification of parallel processor is constructed?

b) Give ONE (1) example for each parallel processor architecture.

QUESTION A2 **(5 marks)**

Artificial intelligence (A1) is the computer ability to exhibit human intelligent behaviors such as smart home applications, robots, intelligent tutoring systems and others.

a) In which generation is A1 introduced?

b) What is the programming language used for this technology?

c) What is the advantage of embedding A1 in our everyday life?

d) A1 can also posed as a threat to community. Discuss ONE (1) possible scenario when A1 can harm people.

QUESTION A3 **(5 marks)**

Given the following Figure 2.

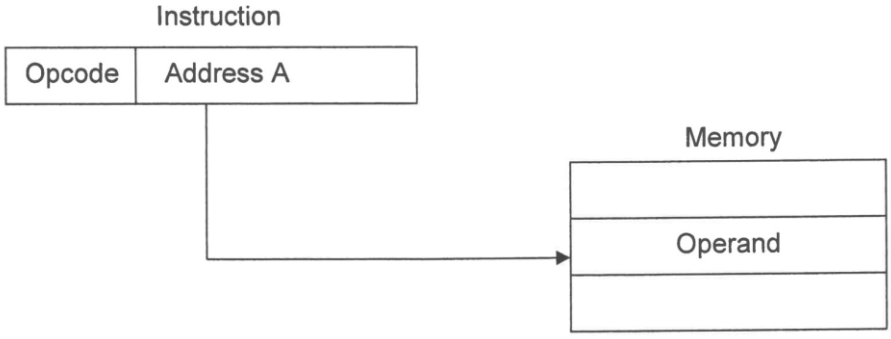


Figure 2

a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

c) List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in 9(a).

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QUESTION A4 **(5 marks)**

Convert 3.25 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

QUESTION A5 **(10 marks)**

Question 5 is based on the following scenario. "Nurathia Farhana browses the UiTM4fly.com website to book for a return flight from KLIA to LAX. While comparing for a better price, two messages popped up in her screen. Figure 1 (a) and (b) illustrate the conveyed messages.

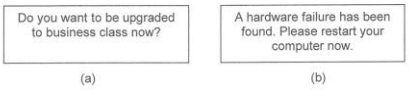


Figure 1: The pop-up messages

(b)

(a)

a) Which message is the non-maskable interrupt? Justify your answer by giving appropriate reason.

b) Differentiate between maskable and non-maskable interrupt?

c) Outline the common way to handle multiple interrupts as happened in the given scenario.

QUESTION A6 **(10 marks)**

Answer both parts for the same expression. Assume an 8-bit byte.
Expression: 63 - (-41)

a) Calculate 63 - (-41) using sign and magnitude.

b) Calculate 63 - (-41) using 1's and 2's complement.

QUESTION A7 **(10 marks)**

a) Differentiate between organization and architecture.

b) Explain the function of THREE (3) Central Processing Unit (CPU) components.

QUESTION A8 **(10 marks)**

a) Discuss TWO (2) functions of an input/output module.

b) External input/output devices can be broadly categorized into three categories. List TWO (2) of the categories and provide an appropriate example of devices for each categories.

c) Name ONE (1) input/output technique that requires active CPU intervention and ONE (1) technique that does not require CPU intervention.

QUESTION A9 **(10 marks)**

a) Compare system interconnection and CPU interconnection in terms of:

i. function

ii. physical appearance.

b) Describe the function of the THREE (3) different bus types in a computer system.

QUESTION A10 **(10 marks)**

a) Elaborate THREE (3) types of replacement algorithm.

b) List TWO (2) types of mapping that require a replacement algorithm.

c) Discuss the reason why the replacement algorithm needs to be implemented.

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

$$y = A - \frac{2C}{B + A^2} * D$$

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three- address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

One-address	Two-address	Three-address
LOADp	MOV (p,q)	ADD (p.q.r)
STORp	ADD (p,q)	SUB (p,q,r)
ADDp	SUB (p,q)	MUL (p.q.r)
SUBp	MUL (p,q)	DIV(p,q,r)
MULp	DIV(p,q)	
DIVp		